Coding & Solution

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/[your-username]/Import-Data-Using-Transform-Maps
Programming Language(s)	Python, JavaScript

Framework(s) Used	Django / FastAPI (Backend), React.js (Frontend)
Key Features Implemented	• **Spreadsheet Drag-and-Drop Upload:** Ingests Excel (.xlsx) and CSV spreadsheets into staging tables. • **Dynamic Transform Mapping Engine:** Auto-suggests and maps source columns to target database fields. • **Coalesce Matching Logic:** Prevents duplicate record insertion by enforcing matching key rules. • **Execution Log Dashboard:** Displays real-time status, total rows processed, skipped records, and row-level error reports.
Pending / Incomplete Features	• Automated email notification alerts for failed bulk import sets. • Direct scheduled background cron jobs for periodic automated spreadsheet processing.
Setup / Run Instructions	**1.** Clone repository: git clone https://github.com/[your-username]/Import-Data-Using-Transform-Maps.git **2.** Install dependencies: pip install -r requirements.txt (Backend) & npm install (Frontend) **3.** Configure database connection in .env file **4.** Run backend server: python manage.py runserver **5.** Run frontend server: npm start

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes

2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The codebase strictly follows PEP 8 standards for Python backend scripts and modular component architecture for React.js frontend components. Robust error handling with try-except blocks ensures graceful handling of corrupt file uploads, column header mismatches, and database timeout issues. All source code, configuration files, and database migration scripts have been properly committed to the GitHub repository.